

Physiology lecture 9

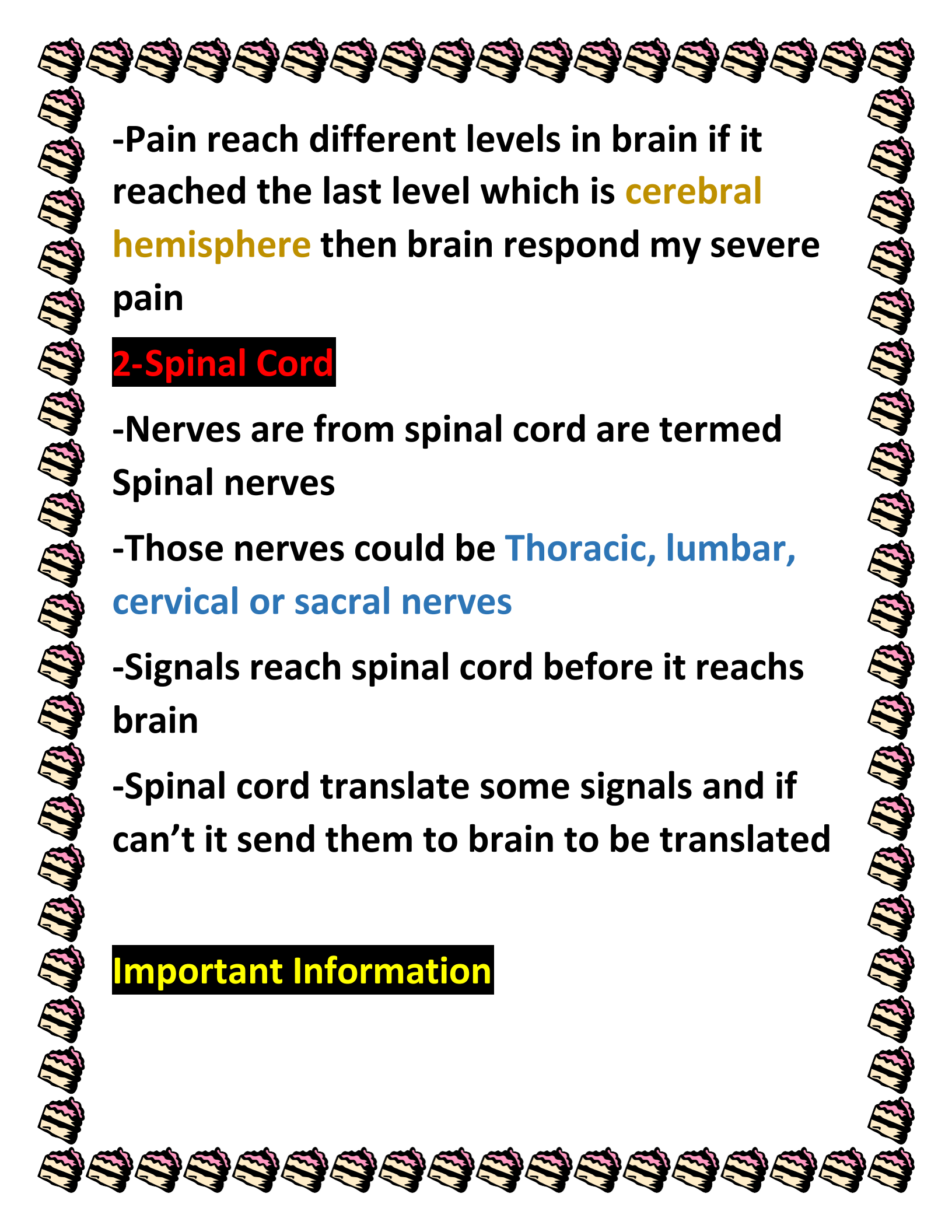
Central Nervous System (CNS)

Brain

Spinal cord

1-Brain

- Nerves are from **BRAIN** are termed Cranial nerves
- Number of nerves: 12
- Note: Nerve 3,7,9,10 has parasympathetic effect and secrete **acetylcholine**
- Protected by Skull and meningeal layers
(**Arachnoid mater, dura mater & PIA mater**)



-Pain reach different levels in brain if it reached the last level which is **cerebral hemisphere** then brain respond my severe pain

2-Spinal Cord

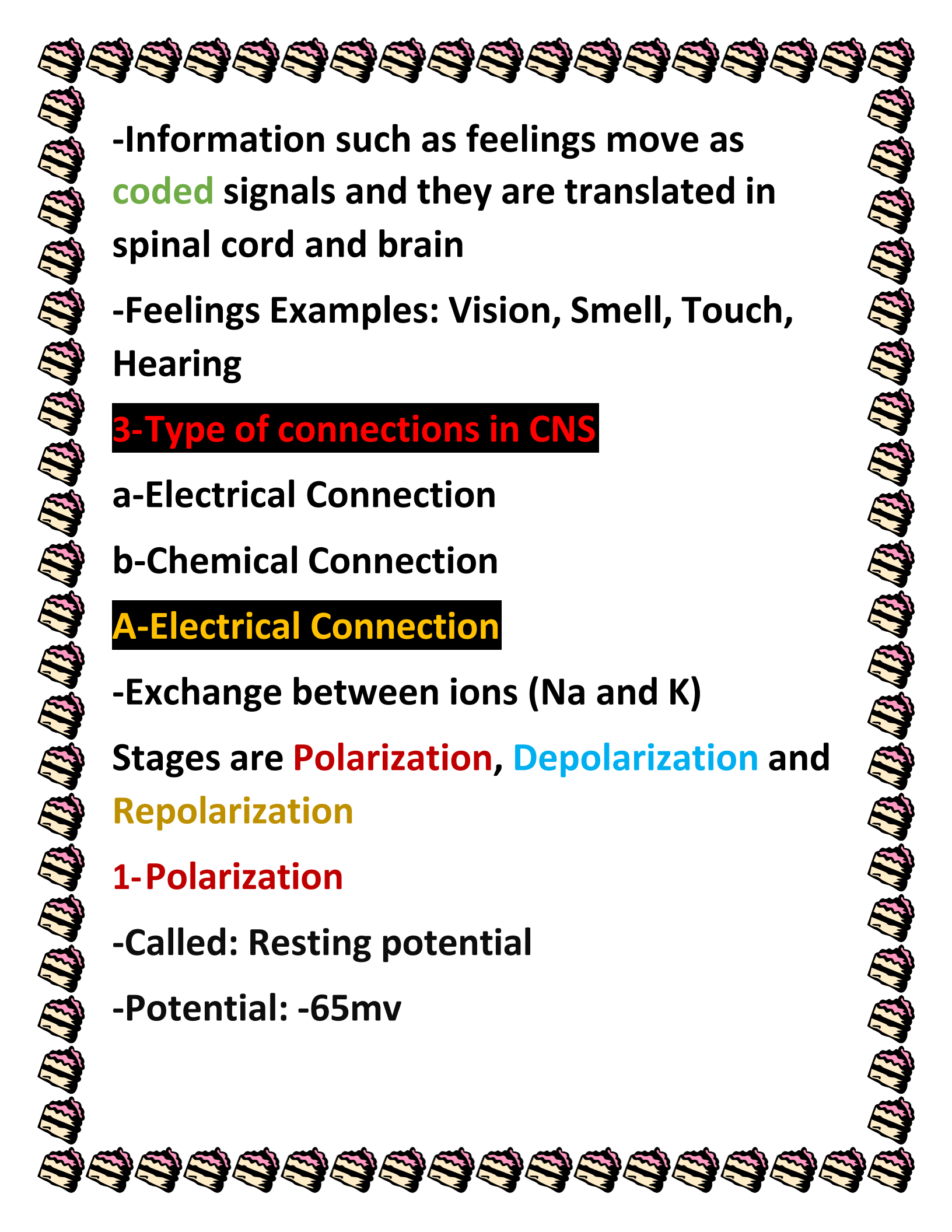
-Nerves are from spinal cord are termed Spinal nerves

-Those nerves could be **Thoracic, lumbar, cervical or sacral nerves**

-Signals reach spinal cord before it reaches brain

-Spinal cord translate some signals and if can't it send them to brain to be translated

Important Information



-Information such as feelings move as **coded** signals and they are translated in spinal cord and brain

-Feelings Examples: Vision, Smell, Touch, Hearing

3-Type of connections in CNS

a-Electrical Connection

b-Chemical Connection

A-Electrical Connection

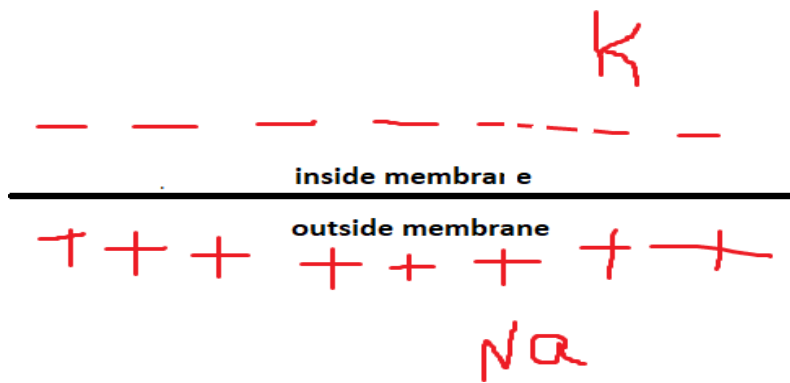
-Exchange between ions (Na and K)

Stages are **Polarization**, **Depolarization** and **Repolarization**

1-Polarization

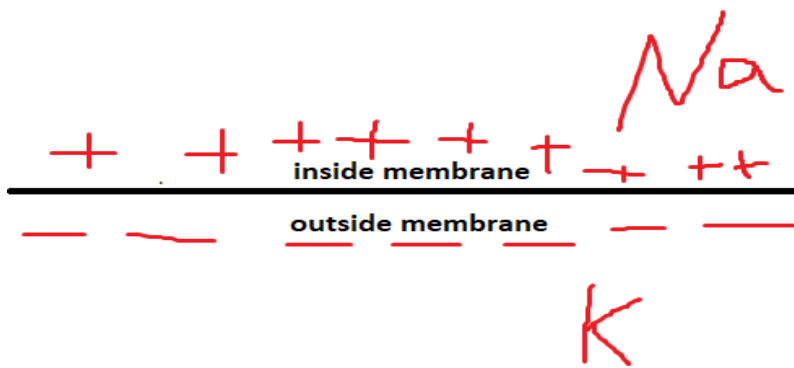
-Called: Resting potential

-Potential: -65mv

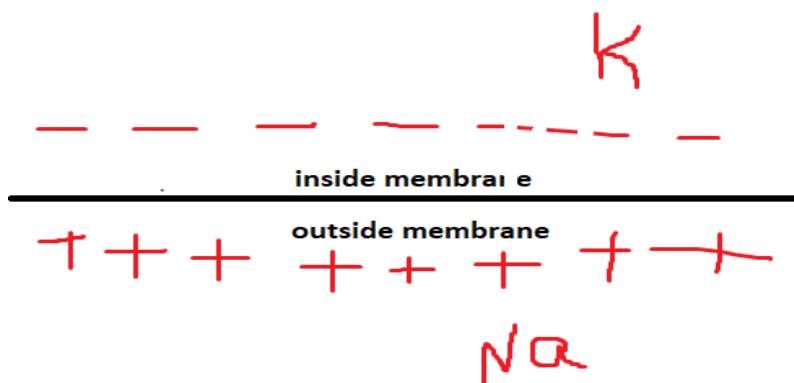


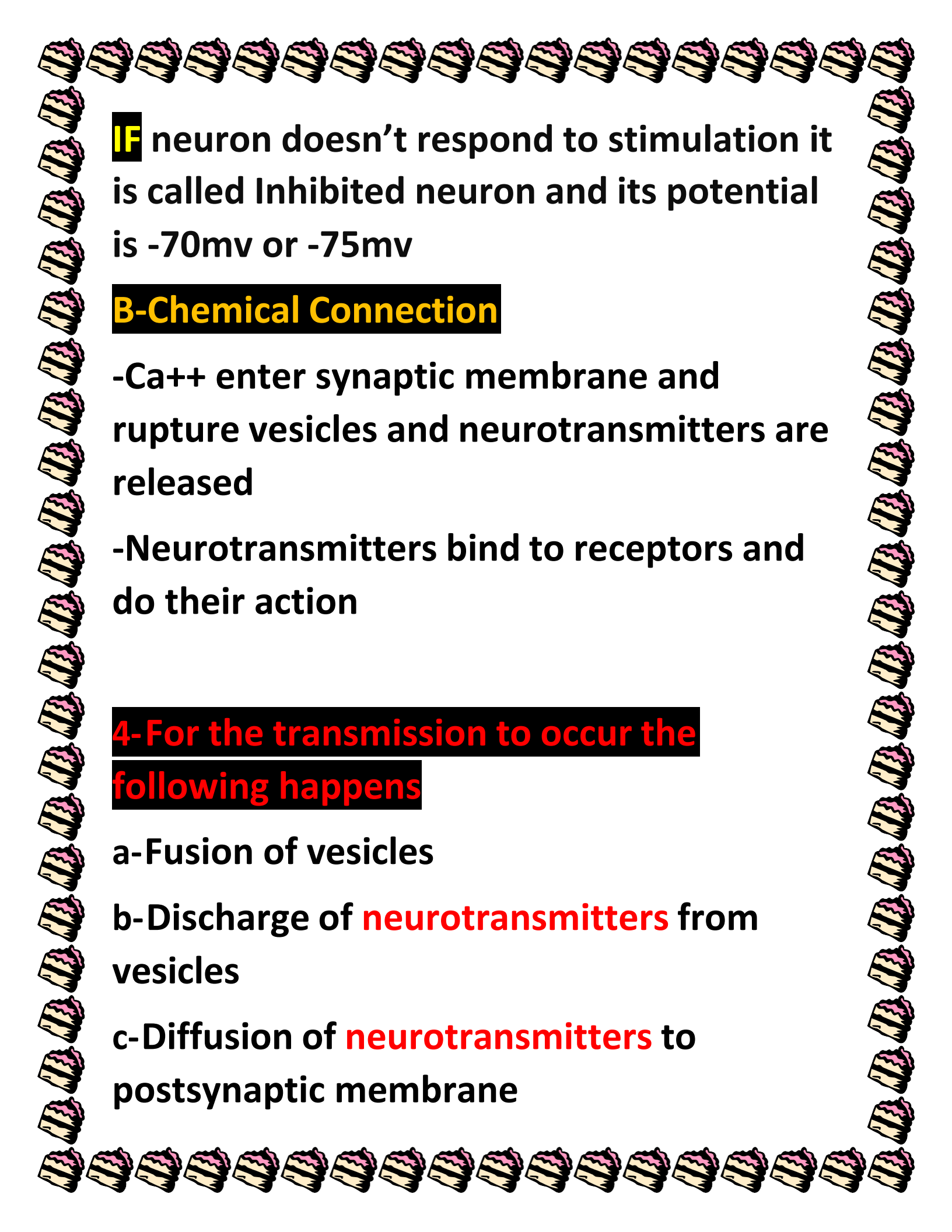
2-Depolarization

- Called: action potential
- Neuron in this case: excited
- Potential: -45mv



3-Repolarization





IF neuron doesn't respond to stimulation it is called Inhibited neuron and its potential is -70mv or -75mv

B-Chemical Connection

- Ca^{++} enter synaptic membrane and rupture vesicles and neurotransmitters are released

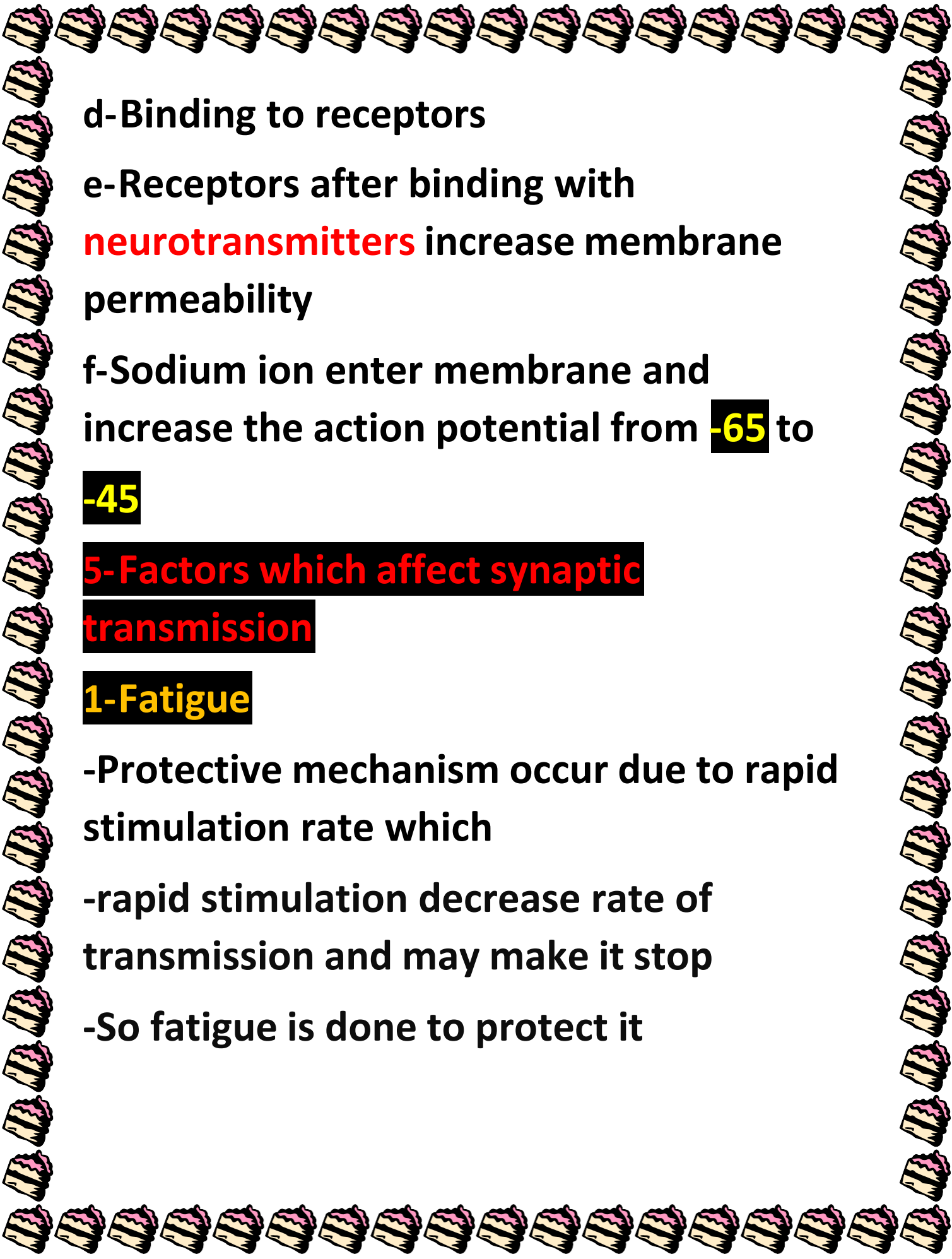
-Neurotransmitters bind to receptors and do their action

4-For the transmission to occur the following happens

a-Fusion of vesicles

b-Discharge of **neurotransmitters** from vesicles

c-Diffusion of **neurotransmitters** to postsynaptic membrane



d-Binding to receptors

e-Receptors after binding with **neurotransmitters** increase membrane permeability

f-Sodium ion enter membrane and increase the action potential from **-65** to **-45**

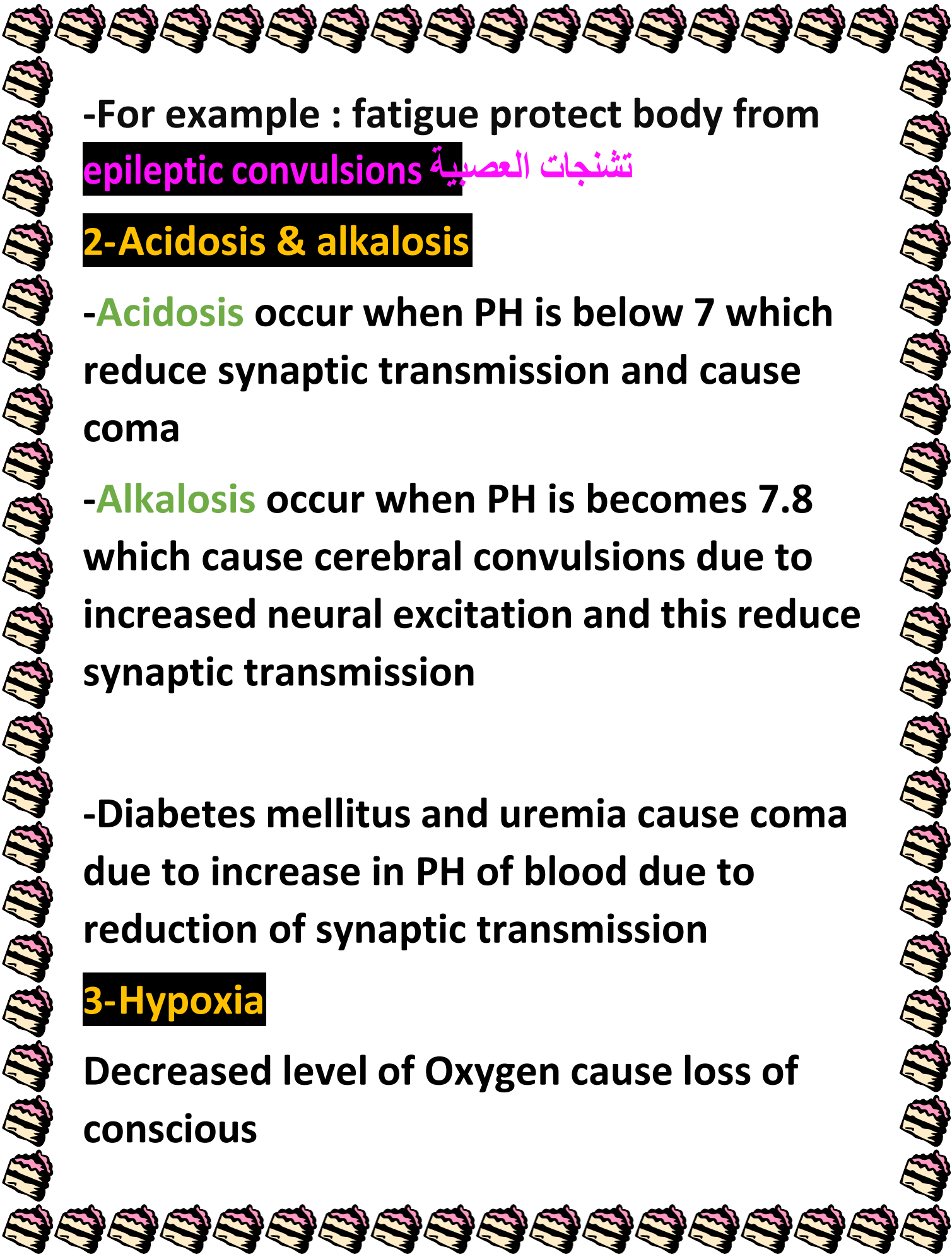
5-Factors which affect synaptic transmission

1-Fatigue

-Protective mechanism occur due to rapid stimulation rate which

-rapid stimulation decrease rate of transmission and may make it stop

-So fatigue is done to protect it



-For example : fatigue protect body from
epileptic convulsions تشنجات العصبية

2-Acidosis & alkalosis

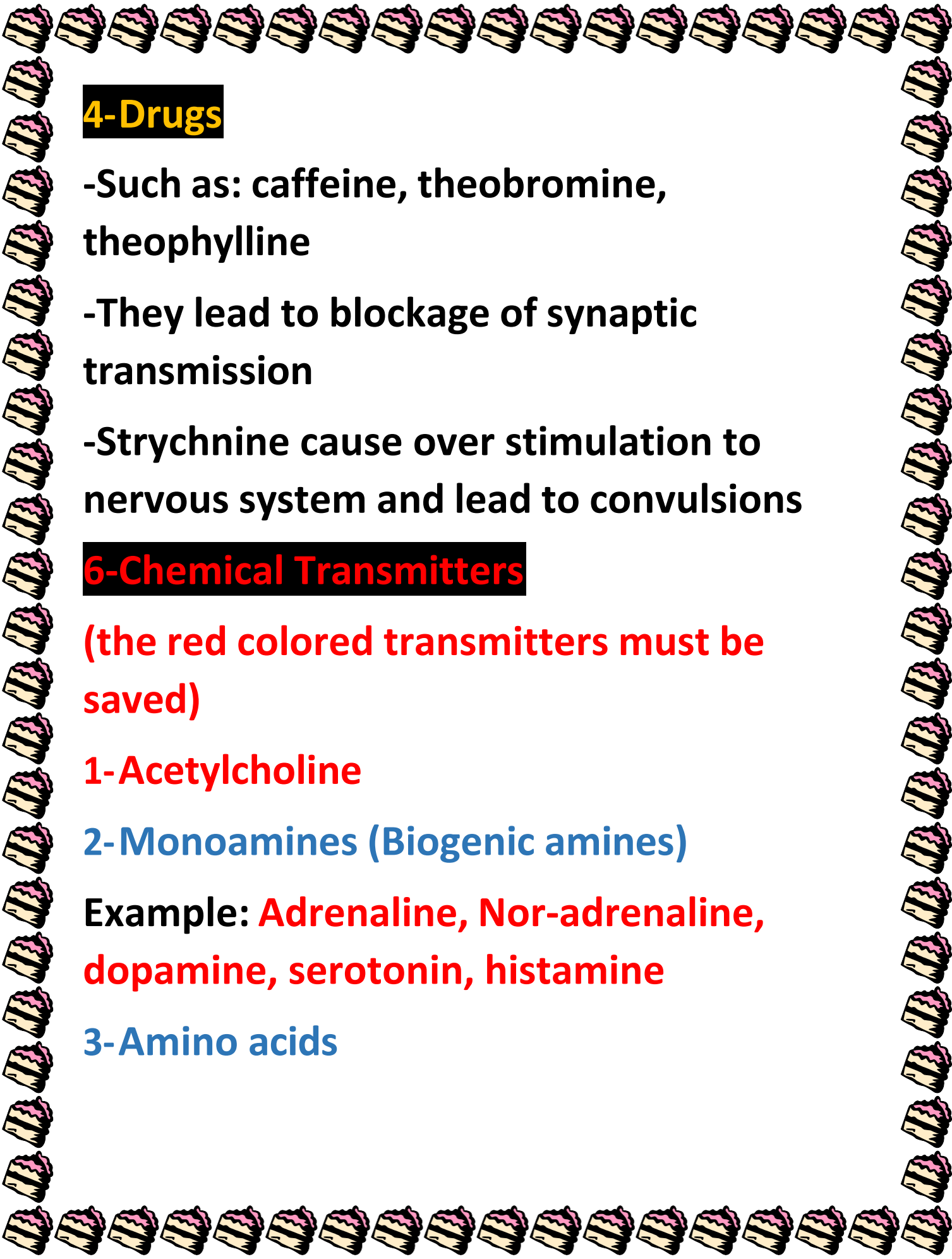
-**Acidosis** occur when PH is below 7 which reduce synaptic transmission and cause coma

-**Alkalosis** occur when PH is becomes 7.8 which cause cerebral convulsions due to increased neural excitation and this reduce synaptic transmission

-Diabetes mellitus and uremia cause coma due to increase in PH of blood due to reduction of synaptic transmission

3-Hypoxia

Decreased level of Oxygen cause loss of conscious



4-Drugs

- Such as: caffeine, theobromine, theophylline
- They lead to blockage of synaptic transmission
- Strychnine cause over stimulation to nervous system and lead to convulsions

6-Chemical Transmitters

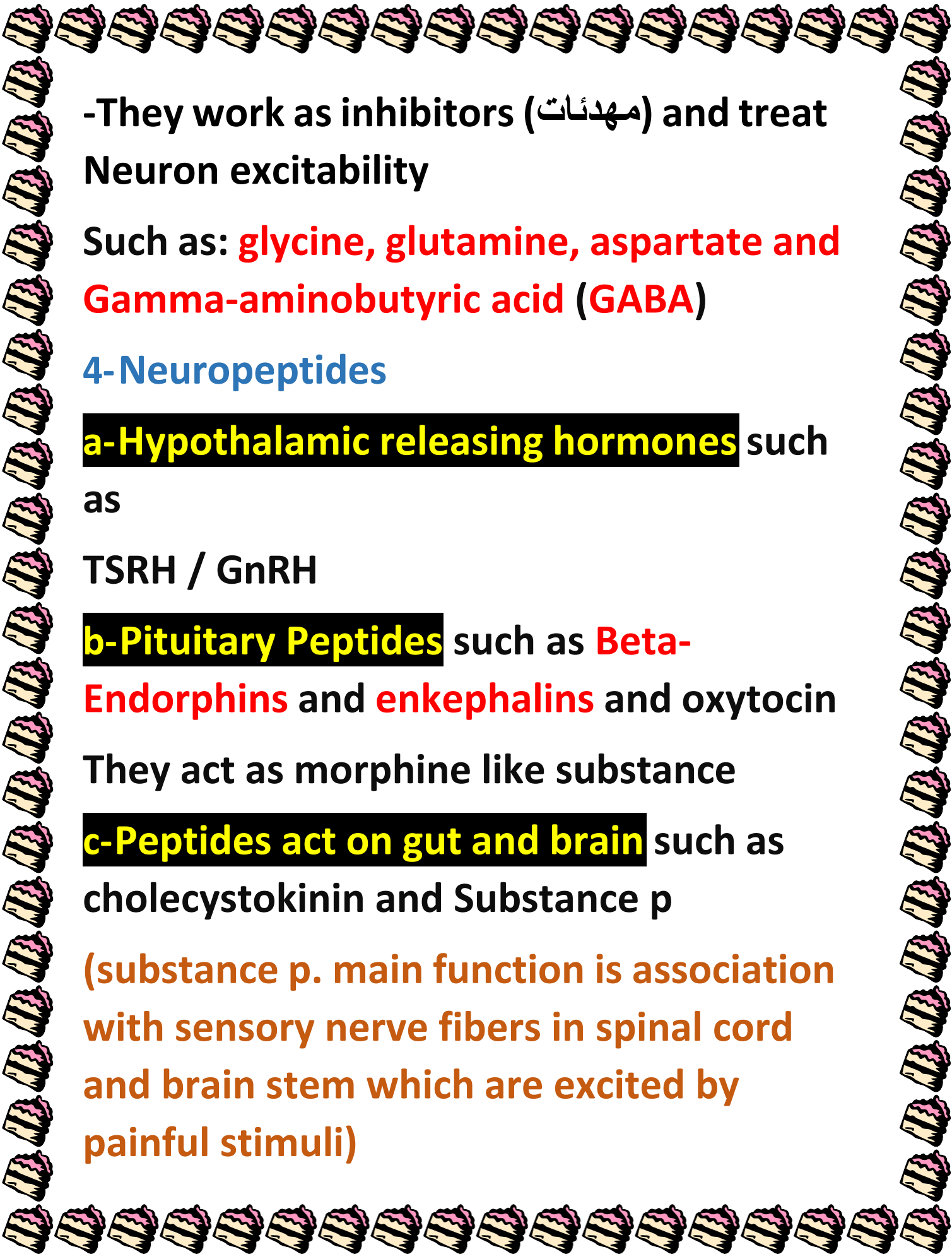
(the red colored transmitters must be saved)

1-Acetylcholine

2-Monoamines (Biogenic amines)

Example: Adrenaline, Nor-adrenaline, dopamine, serotonin, histamine

3-Amino acids



-They work as inhibitors (مهدئات) and treat Neuron excitability

Such as: **glycine, glutamine, aspartate and Gamma-aminobutyric acid (GABA)**

4-Neuropeptides

a-Hypothalamic releasing hormones such as

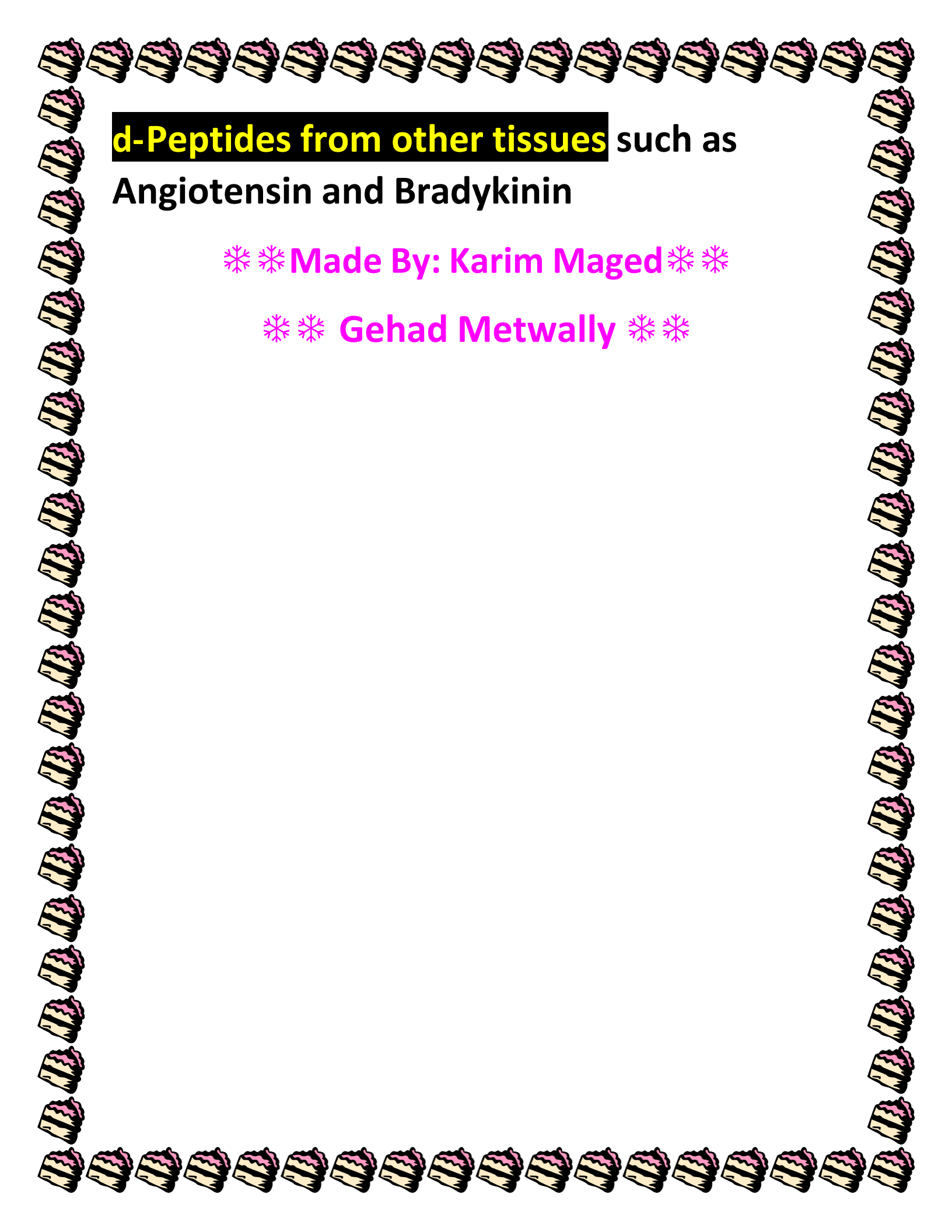
TSRH / GnRH

b-Pituitary Peptides such as **Beta-Endorphins** and **enkephalins** and oxytocin

They act as morphine like substance

c-Peptides act on gut and brain such as cholecystokinin and Substance p

(substance p. main function is association with sensory nerve fibers in spinal cord and brain stem which are excited by painful stimuli)



d-Peptides from other tissues such as
Angiotensin and Bradykinin

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